

Tae-Geun Kim

Postdoctoral Researcher

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Education

Ph.D. in Theoretical Particle Physics, *Yonsei University, Seoul, Republic of Korea* Mar. 2017 – Aug. 2025
Thesis: *Intercommunication between Artificial Intelligence and Physics* (Supervisor: Prof. Seong Chan Park)

B.Sc. in Astronomy, *Yonsei University, Seoul, Republic of Korea* Mar. 2012 – Feb. 2017

Military Service

Technical Research Personnel, *Republic of Korea Army (ROKA)* Sep. 2022 - Aug. 2025

Research Areas & Expertise

- **Dark matter physics**
 - Phenomenology of axion-like particles & primordial black holes
 - Detectability studies of dark matter candidates using various astrophysical and cosmological probes
 - Theoretical modeling and simulation of dark matter interactions
- **Machine Learning for Physics**
 - Development of deep learning models for missing information search and anomaly detection in high-energy physics data
 - Neural network approaches to learn and emulate complex physical systems and dynamics
 - Functional and operator learning for solving differential equations and modeling physical phenomena
- **Scientific & High Performance Computing**
 - Design and implementation of efficient numerical algorithms for physics simulations
 - Optimization of computational methods for large-scale data analysis in astrophysics and particle physics
 - Development of high-performance software tools for scientific computing, with a focus on Rust-based solutions
 - Application of parallel computing techniques to accelerate physics computations

Programming Skills & Tools

- **Primary Languages:** Rust, Python, C++, Julia
- **Frameworks & Libraries**
 - **Numerical Computing:** peroxide, numpy, scipy, pandas/polars, BLAS/LAPACK, eigen, matlab, mathematica
 - **Machine Learning:** pytorch, jax/equinox/optax, wandb, optuna, candle, tensorflow, scikit-learn
 - **Visualization:** matplotlib, vegas, ggplot2, plotly
 - **High Energy Physics:** BlackHawk, galprop, madgraph, root
 - **Quantum Computing:** pennylane, qiskit, cirq, rustqip
 - **Web:** django, vue, firebase, hugo, zola, elm

Honors & Fellowships

- Shanghai Superpostdoc Fellowship, Shanghai Municipal Government (2025-2027)
- Fudan Superpostdoc Fellowship, Fudan University (2025-2027)
- Academy Research Fellowship, Yonsei University (2022-2023)
- Best Oral Presentation Award, KPS 70th Anniversary and 2022 Fall Meeting (2022)

Publications

- 2026 *“Primordial Black Holes as a Factory of Axions: Extragalactic Photons from Axions”*, T.-G. Kim, J.-C. Park, S. C. Park, Y. Park, *Prog. Theor. Exp. Phys. ptag011*, [2212.11977]
- 2025 *“Learning Hamiltonian Dynamics with Bayesian Data Assimilation”*, T. Kim, T.-G. Kim, A. Girard, I. Kolmanovsky, [2501.18808]
- 2024 *“Neural Hamilton: Can A.I. Understand Hamiltonian Mechanics?”*, T.-G. Kim, S. C. Park, [2410.20951]
- 2024 *“HyperbolicLR: Epoch insensitive learning rate scheduler”*, T.-G. Kim, [2407.15200]
- 2024 *“Unsupervised sequence-to-sequence learning for automatic signal quality assessment in multi-channel electrical impedance-based hemodynamic monitoring”*, C. M. Hyun, T.-G. Kim, K. Lee, *Comput. Meth. Prog. Bio.* 108079, [2305.09368]
- 2023 *“DeeLeMa: Missing information search with Deep Learning for Mass estimation”*, K. Ban, D. W. Kang, T.-G. Kim, S. C. Park, Y. Park, *Phys. Rev. Res.* 5. 043186, [2212.12836]

Talks

- 2026 *“Accelerating PBH Phenomenology via Neural Operators”* at *The 2nd AI+HEP in East Asia @ KEK*, Tsukuba, Japan
- 2025 *“From Primordial Black Holes to Nuclei: Solving Inverse Problems in Physics with Operator Learning”* at *Nuclear Lunch Seminar @ Fudan University*, Shanghai, China
- 2025 *“Can AI Understand Hamiltonian Mechanics?”* at *29th International Summer Institute on Phenomenology of Elementary Particle Physics and Cosmology*, Yeosu, Korea
- 2025 *“A Neural Operator for Primordial Black Hole Physics”* at *The 4th workshop on Symmetry and Structure of the Universe*, Jeonju, Korea
- 2025 *“PBH Phenomenology with Operator Learning”* at *LSSU Seminar @ JBNU*, Jeonju, Korea
- 2025 *“AI with Hamiltonian Mechanics: From Predictions to Understanding”* at *AI+HEP in East Asia 2025*, Daejeon, Korea
- 2025 *“Can A.I. Understand Hamiltonian Mechanics?”* at *DEEP-IN Seminar @ RIKEN*, Online
- 2025 *“Can A.I. Understand Hamiltonian Mechanics?”* at *Dark Matter as a portal to New Physics 2025*, Pohang, Korea
- 2025 *“Can A.I. Understand Hamiltonian Mechanics?”* at *Focused workshop on AI in High Energy Physics 2025*, Seoul, Korea
- 2024 *“Can A.I. Understand Hamiltonian Mechanics?”* at *2024 Korea-France STAR Workshop*, Seoul, Korea
- 2024 *“Can A.I. Understand Hamiltonian Mechanics?”* at *KIAS*, Seoul, Korea
- 2024 *“Can A.I. Understand Hamiltonian Mechanics?”* at *21st Saga-Yonsei Joint Workshop*, Seoul, Korea
- 2024 *“Primordial Black Hole dominant Axion background”* at *2024 KPS Fall Meeting*, Yeosu, Korea
- 2023 *“Exploration of PBHs and ALPs through a novel decay model on cosmological scale”* at *27th International Summer Institute on Phenomenology of Elementary Particle Physics and Cosmology*, Nantou, Taiwan

- 2022 *“Exploration of PBHs and ALPs through a novel decay model on cosmological scale”* at 16th International Conference on Interconnections between Particle Physics and Cosmology, Daejeon, Korea
- 2022 *“Constraining ALPs via PBH with time-varying decay process”* at Workshop on Physics of Dark Cosmos: dark matter, dark energy, and all, Busan, Korea
- 2022 *“Constraining ALPs via PBH with time-varying decay process Part.2”* at KPS 70th Anniversary and 2022 Fall Meeting, Busan, Korea

Teaching Experience

2025	Spring	Lecturer	Quantum Universe - Special Research , PHY 9901, Yonsei University, Seoul, Republic of Korea (Lecture about Deep Learning for Physics)
2021	Spring	TA	Introduction to General Relativity , PHY4208, Yonsei University, Seoul, Republic of Korea
2020	Fall	TA	Quantum Mechanics (2) , PHY3102, Yonsei University, Seoul, Republic of Korea
	Spring	TA	Quantum Mechanics (1) , PHY3101, Yonsei University, Seoul, Republic of Korea
2019	Fall	TA	Quantum Mechanics (2) , PHY3102, Yonsei University, Seoul, Republic of Korea
	Spring	TA	Elementary Particle Physics I , PHY8050, Yonsei University, Seoul, Republic of Korea
2017	Fall	TA	Gravity I: General Relativity , PHY8030, Yonsei University, Seoul, Republic of Korea
	Spring	TA	Mathematical Physics (1) , PHY4205, Yonsei University, Seoul, Republic of Korea

Selected Open Source Projects

Peroxide	Sep. 2018 – Present
- Comprehensive Rust numeric library for linear algebra, numerical analysis, and statistics - Customizable features for pure Rust, BLAS/LAPACK integration, and plotting capabilities - Includes automatic differentiation, special functions, DataFrame functionality, and various numerical algorithms	☆ : 665 ⬇ : 1,014,215
Puruspe	Feb. 2020 – Present
- Pure Rust library for special functions with no external dependencies - Implements gamma, beta, and error functions, including their regularized and inverse versions - Lightweight and efficient implementation ideal for mathematical and scientific computing	☆ : 25 ⬇ : 968,211
PyTorch Template	Aug. 2024 – Present
- Flexible PyTorch template for ML experiments with modular structure - Supports YAML-based configuration for easy experiment setup and reproducibility - Supports multiple random seeds, device selection, and learning rate scheduling for robust experimentation	☆ : 10
Quantum Algorithms	Dec. 2023 – Jun. 2024
- Implement quantum algorithms in PennyLane, RustQIP, Qiskit and Cirq - Provide jupyter notebooks for quantum algorithms with detailed descriptions and interactive visualizations	☆ : 7
Radient	Nov. 2023 – Dec. 2023
- Rust library for automatic differentiation using computational graphs - Implements forward and backward propagation for gradient computation - Supports various mathematical operations and provides flexible gradient calculation options	☆ : 4 ⬇ : 3,579

Study Groups

- SYTEARK Machine Learning Study Group** [SYTEARK; MLStudy, ML2022, ML2021] 2019 – Present
- Founder and organizer of a machine learning study group for theoretical physics
 - Current participants: 3 High energy physics graduate students, 1 postdoc and 1 A.I. researcher
 - Weekly rotational presentations on machine learning journals, topics, and personal research
- Yonsei Cosmo-Chalk** [Yonsei_Cosmo-Chalk] 2024 – Mid 2025
- Co-organizer of the weekly blackboard talks in Yonsei HEP-COSMO.
 - Facilitates research presentations, journal discussions, and collaborative dialogues among group members
 - Fosters a supportive environment for intellectual growth and peer-to-peer learning in high-energy physics and cosmology
- ML and QML Study Group** 2021 – Mid 2024
- Participated in study group led by Prof. Myung Hoon Park (Seoultech.) and Prof. Kyoungchul Kong (U. Kansas) at KIAS
 - Study about machine learning and quantum machine learning

Services

- Web master for Yonsei HEP-COSMO** Mar. 2017 – Aug. 2025
- Developed and maintained the [Yonsei HEP-COSMO website](#)
- Beta Reader for Jpub Publishing** Oct. 2019 – Mar. 2025
- Beta reader for Jpub publishing, specializing in error detection for Rust and deep learning publications.
 - Reviewed a total of 16 books as a beta reader.
- Judge for SPARCS Science Hackathon 2024** Oct. 2024
- SPARCS Science Hackathon 2024, “SW for Science”, held by KAIST & Daejeon Metropolitan City
- Web Assistant for the College of Science, Yonsei University** Sep. 2017 – Aug. 2022
- Improved and maintained the [College of Science website](#)